Chapter-3
Review of z-transform

The z-transform is the discrete time analysis of the laplace transform. The z-transform of a discrete-time signal x(n) is defined as the power series,

$$\boxed{X(z) = Z[x(n)] = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}} \quad \text{——— (1)}$$

where z is complex variable.

Eq^n (1) is sometimes called the direct z-transform because it transforms x(n) into complex-plane representation X(z).

The inverse procedure [i.e. obtaining x(n) from X(z)] is called the inverse z-transform i.e.

$$\boxed{x(n) = z^{-1}[X(z)]}$$

The relationship between x(n) & X(z) is indicated by

$$x(n) \xleftrightarrow{z} X(z)$$

Since the z-transform is an infinite power series, it exists only for those values of z for which the series converges. The region of convergence (ROC) of X(z) is the set of all values of z for which X(z) attains a finite value.

Thus, any time we cite a z-transform, we should also indicate its ROC.

Advantages of z-transform:

1. Discrete time signals and LTI systems can be completely characterized by z-transform.
2. The stability of LTI system can be determined using z-transform.
3. By calculating z-transform of given signal, DFT and FT can be determined.
4. The solution of differential equations can be simplified using z-transform.
5. Mathematical calculations are reduced using z-transform. E.g. Convolution operation in transformed into simple multiplication operation.
6. Entire family of digital filters can be obtained from one proto-type design using z-transform.

2017-fall 3.a)

2) one sided z-transform:

A single side z-transform of discrete time signal $x(n)$ may be defined as

$$X^{+}(z) = \sum_{n=0}^{\infty} x(n)\, z^{-n}$$

OR

$$X(z) = \sum_{n=-\infty}^{0} x(n)\, z^{-n}$$

which limits the summation from 0 to ∞. while expanding we put only positive values of n (i.e $n=0$ to $n=\infty$). Due to this, the one-sided z-transform has the following characteristics,

i) It does not contain any information about the signal $x(n)$ for negative values of time (i.e for $n<0$)

ii) It is unique only for causal signals, because only these signals are zero for $n<0$.

iii) The one sided z-transform $X^+(z)$ of $x(n)$ is identical to two-sided z-transform of the signal $x(n)\,u(n)$, the ROC of $X^+(z)$ is always the exterior of a circle.

b) Two-sided z-transform:

A double sided z-transform of discrete-time signal $x(n)$ may be defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

while expanding the summation, we shall put both positive & negative values of 'n'.

Region of Convergence (ROC)

The set of values of z in the z-plane for which the magnitude of $X(z)$ is finite is called the region of convergence (ROC)

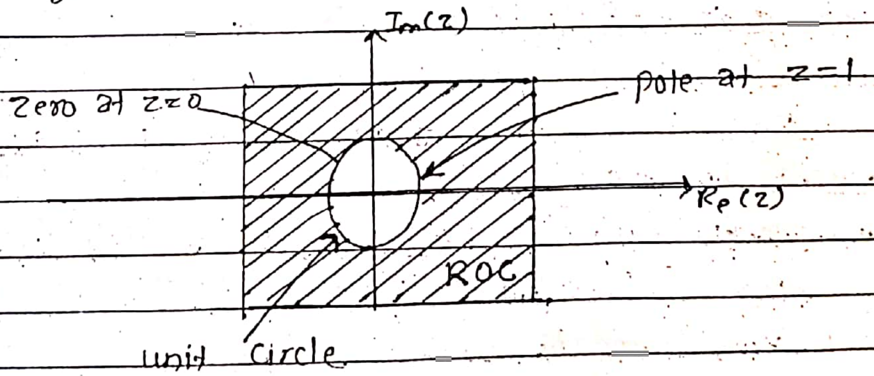

fig: pole-zero plot and ROC of unit step discrete time signal $u(n)$

properties of ROC:

1. The ROC of $X(z)$ consists of a ring circle in the z-plane centered about the origin.

2. The ROC does not contain any pole.

3. The z-transform $X(z)$ converges uniformly if and only if the ROC of $X(z)$ includes the unit circle.
i.e. ROC of $X(z)$ has value of z for which

$$\sum_{n=-\infty}^{\infty} |x(n)\, r^{-n}| < \infty$$

4. If $x(n)$ is a finite duration, then the ROC will be the entire z-plane except $z=0$ and $z=\infty$.

5. If $x(n)$ is Right sided sequence, then ROC will not include infinity.

6. If $x(n)$ is left-sided sequence, then ROC will not include $z=0$. But in a case if $x(n)=0$ for all $n>0$, then the ROC will include $z=0$.

7. If $x(n)$ is a two sided sequence, then ROC will ~~and~~ ~~include~~ consists of a ring in z-plane that includes the circle $|z|=r_0$ i.e. ROC will include the intersection of ROC's of components.

8. If $X(z)$ is rational, then ROC will extend to infinity.

9. If $x(n)$ is causal, then ROC will include $z=\infty$.

10. If $x(n)$ is anti-causal, then ROC will include $z=0$.

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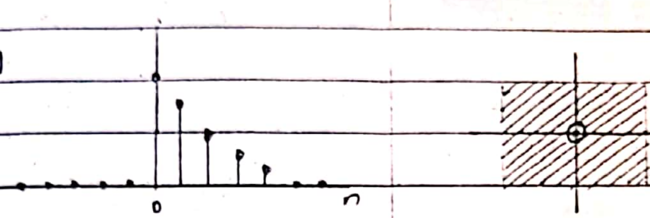
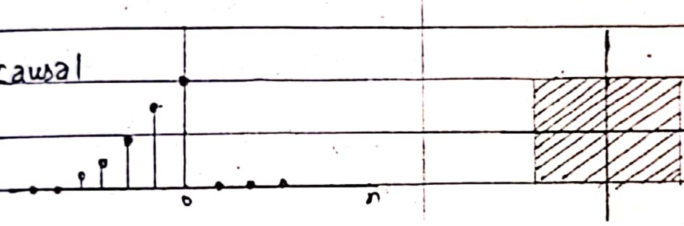
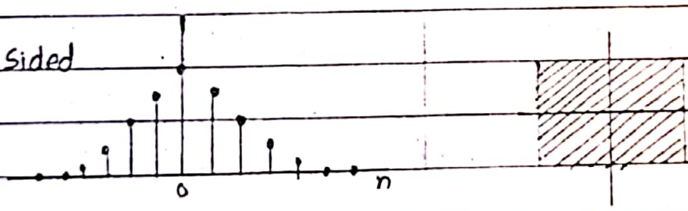
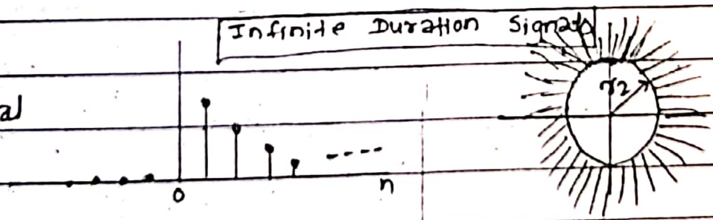
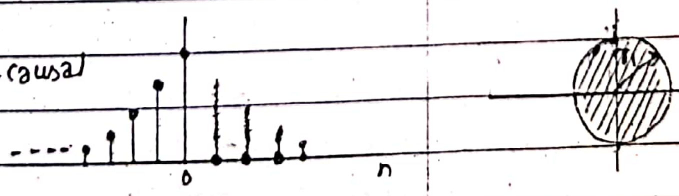
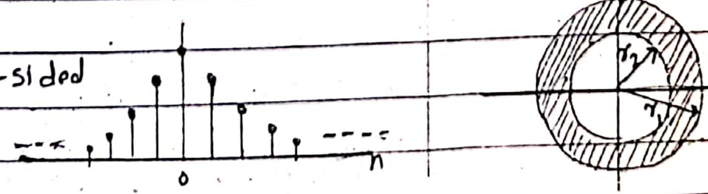
Signal		ROC
Finite - Duration Signals		
Causal		Entire z-plane except $z = 0$
Anticausal		Entire z-plane except $z = \infty$ and
Two-Sided		Entire z-plane except $z = 0$ and $z = \infty$
Infinite Duration Signals		
Causal		$\|z\| > r_2$
Anti-causal		$\|z\| < r_1$
Two-sided		$r_2 \le \|z\| < r_1$

properties of z-transform:

1. Linearity:

If $x_1(n) \xleftrightarrow{z} X_1(z)$

and $x_2(n) \xleftrightarrow{z} X_2(z)$ then

$$x(n) = a_1 x_1(n) + a_2 x_2(n) \xleftrightarrow{z} X(z) = a_1 X_1(z) + a_2 X_2(z)$$

where
a_1 & a_2 are any constant.

2. Time Shifting:

If $x(n) \xleftrightarrow{z} X(z)$ then

$$x(n-k) \xleftrightarrow{z} z^{-k} X(z)$$

ROC of $z^{-k} X(z)$ is the same as that of $X(z)$ except for $z=0$ if $k>0$ and $z=\infty$ if $k<0$.

3. Scaling in z-domain:

If $x(n) \xleftrightarrow{z} X(z)$ ROC: $r_1 < |z| < r_2$

then,

$$a^n x(n) \xleftrightarrow{z} X(a^{-1}z) \quad \text{ROC: } |a| r_1 < |z| < |a| r_2$$

where, a is any constant may be real or complex

4. Time reversal: (affect ROC)

If $x(n) \xleftrightarrow{z} X(z)$ ROC: $r_1 < |z| < r_2$

then

$$x(-n) \xleftrightarrow{z} X(z^{-1}) \quad \text{ROC: } \frac{1}{r_2} < |z| < \frac{1}{r_1}$$

5. Differentiation in z-domain:

If $x(n) \xleftrightarrow{z} X(z)$ } same ROC.

then,

$$n\,x(n) \xleftrightarrow{z} -z\frac{dX(z)}{dz}$$

or $$n\,x(n) \xleftrightarrow{z} z^{-1}\frac{dX(z)}{dz^{-1}}$$

6. Convolution of two sequences:

If $x_1(n) \xleftrightarrow{z} X_1(z)$

and

$x_2(n) \xleftrightarrow{z} X_2(z)$ then,

$$x(n) = x_1(n) * x_2(n) \xleftrightarrow{z} X(z) = X_1(z)\,X_2(z)$$

7. Multiplication of two sequences:

If $x_1(n) \xleftrightarrow{z} X_1(z)$

and

$x_2(n) \xleftrightarrow{z} X_2(z)$ then

$$x(n) = x_1(n)\,x_2(n) \xleftrightarrow{z} X(z) = \frac{1}{2\pi j}\oint_c X_1(v)\,X_2\left(\frac{z}{v}\right)v^{-1}dv$$

8. Correlation of two sequences:

If $x_1(n) \xleftrightarrow{z} X_1(z)$

and

$x_2(n) \xleftrightarrow{z} X_2(z)$ then,

$$r_{x_1,x_2}(l) = \sum_{n=-\infty}^{\infty} x_1(n)\,x_2(n-l) \xleftrightarrow{z} R_{x_1x_2}(z) = X_1(z)\,X_2(z^{-1})$$

9. Initial value Theorem:

If $x(n)$ is causal [i.e. $x(n)=0$ for $n<0$] then initial value of $x(n)$ is

$$x(0) = \lim_{n \to 0} x(n) = \lim_{|z| \to \infty} X(z)$$

10. Final value Theorem

For a discrete time signal $x(n)$, if $X(z)$ and poles of $X(z)$ are inside the unit circle, then the final value of $x(n)$ is

$$x(\infty) = \lim_{n \to \infty} x(n) = \lim_{|z| \to 1} \left[(1-z^{-1}) X(z)\right]$$

11. Parseval's Theorem:

If $x_1(n)$ and $x_2(n)$ are complex-valued sequences, then

$$\sum_{n=-\infty}^{\infty} x_1(n)\, x_2^*(n) = \frac{1}{2\pi j} \oint_c X_1(v)\, X_2^*\left(\frac{1}{v^*}\right) v^{-1}\, dv$$

provided that $r_{1l}\, r_{2l} < 1 < r_{1u}\, r_{2u}$, where, $r_{1l} < |z| < r_{1u}$ and $r_{2l} < |z| < r_{2u}$ are ROC of $X_1(z)$ and $X_2(z)$.

Inverse z-transform:

The procedure for transforming from z-domain to the time domain is called the inverse z-transform.

Mathematically,

$$x(n) = z^{-1}[X(z)]$$

To perform inverse z-transform, any on of these methods are used

i) Long division method / power series expansion

ii) Partial fraction expansion

iii) Residue method.

i) Long division Method:

The z- transform of $x(n)$ is

$$X(z) = \sum_{n=0}^{\infty} x(n)\, z^{-n}$$

which may be expressed as the ratio of two polynomials in z.

$$X(z) = \frac{N(z)}{D(z)}$$

The above ratio of polynomials for z-transform may be divided out to produce a power Series in the form of an equation with the coefficients representing the Sequence values in the time domain as

$$X(z) = \frac{N(z)}{D(z)} = \sum_{n=0}^{\infty} a_n z^{-n} = a_0 z^0 + a_1 z^{-1} + a_2 z^{-2} + \cdots$$

Where coeff. a_n are the values of $x(n)$

2. Partial fraction method:

With the help of partial fraction method, we convert system transfer function into a sum of standard functions. To do this, firstly denominator of the transfer function $H(z)$ is factorized into prime factors and then simple poles of $H(z)$ is obtained.

$H(z)$ is given by

$$H(z) = A_0 + \frac{A_1}{z-P_1} + \frac{A_2}{z-P_2} + \cdots + \frac{A_N}{(z-P_N)}$$

3. Residue method:

In this method we obtain inverse z-transform $x(n)$, by summing residues of $[X(z)\, z^{n-1}]$ at all poles.

Mathematically

$$x(n) = \sum_{\substack{\text{all poles}\\ X(z)}} \text{residues of } [X(z)\, z^{n-1}]$$

The residue for any pole of order m at $z=\beta$ is

$$\text{Residue} = \frac{1}{(m-1)!} \lim_{z\to\beta} \left\{ \frac{d^{m-1}}{dz^{m-1}} \left[(z-\beta)^m X(z) \cdot z^{n-1}\right] \right.$$

	Signal $x(n)$	z-transform $X(z)$	ROC
1.	$\delta(n)$	1	Entire z plane
2.	$\delta(n-k)$	z^{-k}	Entire z-plane except $z=0$.
3.	$\delta(n+k)$	z^{k}	Entire z-plane except $z=\infty$
4.	$u(n)$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$\lvert z\rvert > 1$
5.	$u(-n)$	$\frac{z^{-1}}{z^{-1}-1}$	$\lvert z\rvert > 1$
6.	$n\,u(n)$	$\frac{z}{(z-1)^2}$	$\lvert z\rvert > 1$
7.	$a^n u(n)$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$

	Signal	z-transform	ROC.
8.	$n a^n u(n)$	$\frac{a z^{-1}}{(1-a z^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
9.	$-a^n u(-n-1)$	$\frac{1}{1-a z^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
10.	$-n a^n u(-n-1)$	$\frac{a z^{-1}}{(1-a z^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
11.	$(\cos \omega_0 n) u(n)$	$\frac{1 - z^{-1}\cos\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}}$	$\lvert z\rvert > 1$
12.	$(\sin \omega_0 n) u(n)$	$\frac{z^{-1}\sin\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}}$	$\lvert z\rvert > 1$
13.	$(a^n \cos\omega_0 n) u(n)$	$\frac{1 - a z^{-1}\cos\omega_0}{1 - 2a z^{-1}\cos\omega_0 + a^{2} z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$
14.	$(a^n \sin\omega_0 n) u(n)$	$\frac{a z^{-1}\sin\omega_0}{1 - 2a z^{-1}\cos\omega_0 + a^2 z^{-2}}$	$\lvert z\rvert > \lvert a\rvert$

Additional property of one side z-transform: (Shifting)

1. Time delay

If $x(n) \xleftrightarrow{z^+} X^+(z)$ then

$$x(n-k) \xleftrightarrow{z^+} z^{-k}\left[X^+(z) + \sum_{n=1}^{k} x(-n) z^{n}\right], \quad k > 0.$$

In case, $x(n)$ is causal, then

$$x(n-k) \xleftrightarrow{z^+} z^{-k} X^+(z)$$

proof:

$$X^+(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$z^+\{x(n-k)\} = z^{-k}\left[\sum_{l=-k}^{-1} x(l) z^{-l} + \sum_{l=0}^{\infty} x(l) z^{-l}\right]$$

$$= z^{-k}\left[\sum_{l=-1}^{-k} x(l) z^{-l} + X^+(z)\right]$$

2. Time advance:

If $x(n) \xleftrightarrow{z^+} X^+(z)$ then

$$x(n+k) \xleftrightarrow{z^+} z^{k}\left[X^+(z) - \sum_{n=0}^{k-1} x(n) \cdot z^{-n}\right], \quad k > 0 \quad (2)$$

proof:

$$z^+\{x(n+k)\} = \sum_{n=0}^{\infty} x(n+k) z^{-n} = z^{k} \cdot \sum_{l=k}^{\infty} x(l) z^{-l}$$

where, we have changed the Index of Summation from n to $l = n+k$, then

$$X^+(z) = \sum_{l=0}^{\infty} x(l) z^{-l} = \sum_{l=0}^{k-1} x(l) z^{-l} + \sum_{l=k}^{\infty} x(l) z^{-l}$$

By Combining we get the desired result eqn (2).

~~find the inverse~~

Q $x(n) = \{1, 2, 3\}$ & $y(n) = \{3, 4, 8, 8, 3\}$, $h(n) = ?$

$X(z) = 1 + 2z^{-1} + 3z^{-2}$ & $Y(z) = 3 + 4z^{-1} + 8z^{-2} + 8z^{-3} + 3z^{-4}$

$\therefore H(z) = \frac{Y(z)}{X(z)}$

$$
\begin{array}{r|l}
 & 3 - 2z^{-1} + 3z^{-2} \\
\hline
1 + 2z^{-1} + 3z^{-2} & 3 + 4z^{-1} + 8z^{-2} + 8z^{-3} + 3z^{-4} \\
 & 3 + 6z^{-1} + 9z^{-2} \\
 & - \quad - \quad - \\
\hline
 & -2z^{-1} - z^{-2} + 8z^{-3} \\
 & -2z^{-1} - 4z^{-2} - 6z^{-3} \\
 & + \quad + \quad + \\
\hline
 & 3z^{-2} + 14z^{-3} + 3z^{-4} \\
 & 3z^{-2} + 6z^{-3} \\
\hline
\end{array}
$$

2014 - spring
2.b) Soln

Q. Determine the causal signal $x(n)$ having $X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})^2}$

Here, $X(z) = \frac{1}{(1-2z^{-1})(1-z^{-1})^2} = \frac{1}{(1-2z^{-1})} \cdot \frac{1}{(1-z^{-1})^2} = X_1(z) \cdot X_2(z)$

Let, $X_1(z) = \frac{1}{1-2z^{-1}} \Rightarrow x_1(n) = (2)^n u(n)$

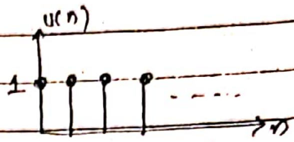
$X_2(z) = \frac{1}{(1-z^{-1})^2} \Rightarrow$

Numericals:

Q. Find the z-transform of $x(n) = u(n)$.

Sol^n:

Here, $x(n) = u(n)$



we know,

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\cdot z^{-n} = \sum_{n=-\infty}^{\infty} u(n)\cdot z^{-n}$$

$$= \sum_{n=-\infty}^{-1} u(n)\cdot z^{-n} + \sum_{n=0}^{\infty} u(n)\cdot z^{-n}$$

$$= 0 + \sum_{n=0}^{\infty} 1\cdot z^{-n}$$

$$= \sum_{n=0}^{\infty} (z^{-1})^{n}$$

$$= (z^{-1})^0 + (z^{-1})^1 + (z^{-1})^2 + (z^{-1})^3 + \cdots$$

$$\therefore X(z) = 1 + \frac{1}{z} + \left(\frac{1}{z}\right)^2 + \left(\frac{1}{z}\right)^3 + \cdots$$

we know,

$$= 1 + a + a^2 + a^3 + \cdots = \frac{1}{1-a}, \; a \neq 1$$

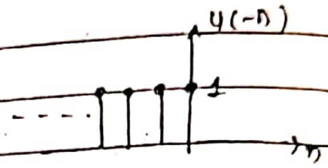
So,

$$X(z) = \frac{1}{1 - \frac{1}{z}} \quad \text{if } \left|\frac{1}{z}\right| < 1 \text{ i.e. } |z^{-1}| < 1$$

$$\text{or, } 1 < |z|$$

$$\therefore X(z) = \frac{z}{z-1} \quad \text{ROC: } |z| > 1$$

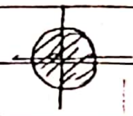
Q. Find $X(z)$ for $x(n) = u(-n)$



Sol^n: Here $x(n) = u(-n)$

we know, $X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$

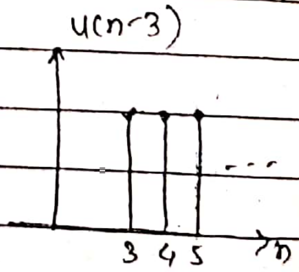
$\therefore X(z) = \sum_{n=-\infty}^{0} 1 \cdot z^{-n} = \sum_{n=-\infty}^{0} (z^{-1})^{n} = \cancel{(z^{-1})^{0}}$



$\therefore X(z) = \sum_{n=0}^{\infty} z^{n} = z^0 + z^1 + z^2 + \cdots = \frac{1}{1-z}$ if $|z| < 1$.

Q. Find $X(z)$ for $x(n) = u(n-3)$

Sol^n: Here, $x(n) = u(n-3)$



we know, $X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

$\therefore X(z) = \sum_{n=3}^{\infty} 1 \cdot z^{-n} = \sum_{n=3}^{\infty} (z^{-1})^{n}$

$\therefore X(z) = (z^{-1})^3 + (z^{-1})^4 + (z^{-1})^5 + \cdots$

$= (z^{-1})^3 \left[1 + (z^{-1}) + (z^{-1})^2 + (z^{-1})^3 + \cdots \right]$

$= (z^{-1})^3 \left[\frac{1}{1-z^{-1}} \right]$ ROC: $|z^{-1}| < 1$ i.e ROC: $|z| > 1$

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Q. Find z-transform of $x(n) = a^n u(n)$, $0 < a < 1$.

Soln: we have, $x(n) = a^n u(n)$

then

$$X(z) = \sum_{n=-\infty}^{\infty} x(n).z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n.1.z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

$$\therefore X(z) = \frac{1}{1-az^{-1}} \quad ROC: |az^{-1}| < 1 \quad \text{i.e } ROC: |z| > |a|$$

Q. Find z-transform of $x(n) = u(n) + u(n-3) + a^n u(n)$.

Soln:- we know, $X(z) = \sum_{n=-\infty}^{\infty} x(n).z^{-n}$

Here, $x(n) = u(n) + u(n-3) + a^n u(n)$

Let $x_1(n) = u(n)$

$x_2(n) = u(n-3)$

$x_3(n) = a^n u(n)$

$$\therefore X_1(z) = \frac{1}{1-z^{-1}}, \quad ROC: |z^{-1}| < 1 \quad \text{i.e. } ROC \ |z| > 1$$

$X_2(z) = (z^{-1})^3 \cdot \frac{1}{1-z^{-1}}$, ROC: $|z^{-1}| < 1$ i.e ROC: $|z| > 1$

$X_3(z) = \frac{1}{1-az^{-1}}$ ROC: $|az^{-1}| < 1$ i.e ROC: $|z| > |a|$

finally,

$$X(z) = X_1(z) + X_2(z) + X_3(z)$$

$$= \frac{1}{1-z^{-1}} + \frac{z^{-3}}{1-z^{-1}} + \frac{1}{1-az^{-1}} \quad \text{ROC: } |z| > |a|$$

2015-fall 3.a) Q. find z-transform of the signal $x(n) = a^n \cos \omega_0 n \, u(n)$

Soln:-

2015-fall 3.b) Q. find z-transform of signal $x(n) = n \cdot a^n u(n)$

Soln:- Here, given $x(n) = n a^n u(n)$ ——(1)

let $x_1(n) = a^n u(n)$

we know,

z-transform of $a^n u(n)$ is $X(z) = \frac{z}{z-a}$ ROC: $|z| > |a|$

so, eqn (1) becomes

$$x(n) = n \, x_1(n)$$

According to differentiation property, we have

$$n\,x_1(n) \xleftrightarrow{z} -z\frac{d\,X_1(z)}{dz} \quad \text{or} \quad z^{-1}\frac{d\,X_1(z)}{dz^{-1}}$$

$$\therefore Z\{n\,a^n u(n)\} = -z\frac{d}{dz}\left(\frac{z}{z-a}\right) \quad \text{or,} \quad z^{-1}\frac{d}{dz^{-1}}\left(\frac{1}{1-az^{-1}}\right)$$

$$= -z\frac{d}{dz}\left(z(z-a)^{-1}\right) \qquad = +z^{-1}\cdot\frac{(-1)\cdot(-a)}{(1-az^{-1})^2}$$

$$= \frac{az^{-1}}{1}$$

$$= -z\left[z\cdot(-1)\cdot(z-a)^{-2} + (z-a)^{-1}\cdot 1\right]$$

$$= -z\left[\frac{-z}{(z-a)^2} + \frac{1}{z-a}\right]$$

$$= -z\left[\frac{-z+(z-a)}{(z-a)^2}\right]$$

$$= -z\left[\frac{-a}{(z-a)^2}\right]$$

$$\therefore Z\{n\,a^n u(n)\} = \frac{az}{(z-a)^2} \qquad \text{ROC: } |z| > |a|$$

OR. $$Z\{n\,a^n u(n)\} = \frac{az^{-1}}{(1-az^{-1})^2} \qquad \text{ROC: } |z| > |a|$$

2016 spring Q.b) Q. Find the z-transform of $x(n) = -a^n u(-n-1)$

Soln: Here given, $x(n) = -a^n u(-n-1)$

we have

$$z\{u(n)\} = \frac{1}{1-z^{-1}} \quad ROC: |z| > 1.$$

Applying time shifting property, we get

$$z\{u(n-1)\} = z^{-1} \times \frac{1}{1-z^{-1}} = \frac{1}{z-1} \quad ROC: |z| > 1$$

Applying scaling property, we get

$$z\, a^n u(n-1) = \frac{1}{(a^{-1}z) - 1} = \frac{a}{z-a} \quad ROC: |z| > a.$$

we know, $z\{u(-n)\} = \frac{z^{-1}}{z^{-1}-1}$ ROC: $|z^{-1}| > 1$

Applying shifting property, we get

$$z\{u(-n-1)\} = z^{+1}\left(\frac{z^{-1}}{z^{-1}-1}\right) = \frac{z}{1-z} \quad ROC: |z| < 1$$

Applying scaling property,

$$z\{a^n u(-n-1)\} = \frac{(a^{-1}z)}{1-(a^{-1}z)} \quad ROC: |a^{-1}z| < 1$$

i.e. ROC: $|z| > |a|$

$$= \frac{z}{a-z} \quad |z| < |a|$$

$$\therefore z\{-a^n u(-n-1)\} = \frac{-z}{a-z} \quad ROC: |z| < |a|$$

Hence, $z\{-a^n u(-n-1)\} = \frac{z}{z-a}$ ROC: $|z| > |a|$

Q. Find the z-tramform of $x(n) = n^2 u(n)$.

Soln: Here, given, $x(n) = n^2 u(n)$

$= n \cdot n\,u(n)$

We know, $z\{u(n)\} = \frac{1}{1-z^{-1}}$ ROC: $|z| > 1$.

using differentiation property,

$$z\{n\,u(n)\} = z^{-1} \frac{d}{dz^{-1}}\left(\frac{1}{1-z^{-1}}\right)$$

$$= z^{-1}(-1)\cdot(1-z^{-1})^{-2}(-1).$$

$$= \frac{z^{-1}}{(1-z^{-1})^2} \quad ROC: |z^{-1}| < 1$$

i.e ROC : $|z| > 1$

$$\therefore Z\{n \cdot u(n)\} = \frac{z^{-1}}{(1-z^{-1})^2} \qquad ROC: |z|>1$$

Again applying differentiation property

$$Z\{n^2 \cdot u(n)\} = z^{-1} \frac{d}{dz^{-1}}\left[\frac{z^{-1}}{(1-z^{-1})^2}\right]$$

$$= z^{-1} \cdot \left[\frac{1 \cdot (1-z^{-1})^2 - (2)(1-z^{-1})^{2-1} \cdot (-1)}{\{(1-z^{-1})^2\}^2}\right]$$

$$\therefore Z\{n^2 u(n)\} = z^{-1}\left[\frac{(1-z^{-1})^2 + 2(1-z^{-1})}{(1-z^{-1})^4}\right]$$

Q. Find the z-transform of $x(n) = \cos \omega_0 n \, u(n)$. $n>0$

Sol^n:- Here, $x(n) = \cos \omega_0 n \, u(n)$

$$= \left[\frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}\right] u(n)$$

$$x(n) = \underbrace{\frac{1}{2} e^{j\omega_0 n} u(n)}_{x_1(n)} + \underbrace{\frac{1}{2} e^{-j\omega_0 n} u(n)}_{x_2(n)}$$

Let $x_1(n) = \frac{1}{2} e^{j\omega_0 n} u(n)$ & $x_2(n) = \frac{1}{2} e^{-j\omega_0 n} u(n)$

$$\therefore X_1(z) = \frac{1}{2} \sum_{n=0}^{\infty} e^{j\omega_0 (n)} \cdot 1 \cdot z^{-n}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} (e^{j\omega_0} z^{-1})^n$$

$$= \frac{1}{2} \left[1 + e^{j\omega_0} z^{-1} + (e^{j\omega_0} z^{-1})^1 + (e^{j\omega_0} z^{-1})^2 + \cdots \right]$$

$$\therefore X_1(z) = \frac{1}{2} \left[\frac{1}{1 - e^{j\omega_0} z^{-1}} \right]$$; ROC: $|e^{j\omega_0} z^{-1}| < 1$

i.e. ROC: $|z| > |e^{j\omega_0}|$

i.e. $|z| > 1$

Similarly,

$$X_2(z) = \frac{1}{2} \frac{1}{1 - e^{-j\omega_0} z^{-1}}$$ ROC: $|z| > |e^{-j\omega_0}|$

i.e. $|z| > 1$

now,

$$X(z) = X_1(z) + X_2(z)$$

$$= \frac{1}{2} \frac{1}{1 - e^{j\omega_0} z^{-1}} + \frac{1}{2} \frac{1}{1 - e^{-j\omega_0} z^{-1}}$$ ROC: $|z|>$

$$= \frac{1}{2} \left[\frac{(1 - e^{-j\omega_0} z^{-1}) + (1 - e^{j\omega_0} z^{-1})}{(1 - e^{j\omega_0} z^{-1})(1 - e^{-j\omega_0} z^{-1})} \right]$$

$$X(z) = \frac{1}{2}\left[\frac{2 - z^{-1}\left(e^{j\omega_0} + e^{-j\omega_0}\right)}{1 - e^{-j\omega_0}z^{-1} - e^{j\omega_0}z^{-1} + e^{0}z^{-2}}\right] \quad \text{ROC: } |z|>1$$

2014-Sp... 3.?

$$= \frac{1}{2}\left[\frac{2 - z^{-1}\cdot 2\cos\omega_0}{1 - z^{-1}\left(e^{j\omega_0} + e^{-j\omega_0}\right) + z^{-2}}\right]$$

$$= \frac{1}{2}\left[\frac{2 - 2z^{-1}\cos\omega_0}{1 - z^{-1}(2\cos\omega_0) + z^{-2}}\right]$$

$$\therefore X(z) = \frac{1 - z^{-1}\cos\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}} \quad |z|>1. \quad \text{— (1)}$$

Similarly, for $x(n) = \sin\omega_0 n\, u(n)$

$$X(z) = \mathcal{X}\{\sin\omega_0 n\, u(n)\} = \frac{1 - z^{-1}\sin\omega_0}{1 - 2z^{-1}\cos\omega_0 + z^{-2}}$$

2015-fall .2b)

Q. Find the z-transform of $x(n) = a^n \cos\omega_0 n\, u(n)$

Sol^n: Applying scaling Property in eq^n (1) we get

$$Z\{a^n \cos\omega_0 n\, u(n)\} = \frac{1 - (a^{-1}z)^{-1}\cos\omega_0}{1 - 2(a^{-1}z)^{-1}\cos\omega_0 + (a^{-1}z)^{-2}}$$

$$\therefore = \frac{1 - az^{-1}\cos\omega_0}{1 - 2az^{-1}\cos\omega_0 + a^2z^{-2}} \quad |z|>|a| \quad \text{— (11)}$$

2014-Spring 3.a)

Q. Find z-transform of $x(n) = n\, a^n \cos\omega_0 n \cdot u(n)$

Sol^n:- we have from eq^n (1),

Applying differentiation property we get

$$Z\{n \cdot a^n \cos\omega_0 n\, u(n)\} = z^{-1} \frac{d}{dz^{-1}} \left\{ \frac{1 - a z^{-1}\cos\omega_0}{1 - 2az^{-1}\cos\omega_0 + a^2 z^{-2}} \right\}$$

$$= z^{-1} \left[\frac{-a\sin\omega_0 (1 - 2az^{-1}\cos\omega_0 + a^2 z^{-2}) - (-2a\cos\omega_0 + 2a^2 z^{-1})(1 - az^{-1})}{(1 - 2az^{-1}\cos\omega_0 + a^2 z^{-2})^2} \right]$$

$=$

2015-Spring 2.b)

Q. A discrete time signal is given by the expression

$x(n) = n\left(-\frac{1}{2}\right)^n u(n) \circledast \left(\frac{1}{4}\right)^{-n} u(-n)$. Find z-transform

Sol^n:- Here, Let, $x_1(n) = n\left(-\frac{1}{2}\right)^n$ & $x_2(n) = \left(\frac{1}{4}\right)^{-n} u(-n)$

- For $x_1(n) := n\left(-\frac{1}{2}\right)^n$

we know,

$$z\{u(n)\} = \frac{1}{1-z^{-1}} \quad \text{ROC: } |z| > 1$$

using differentiation property,

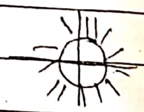
$$z\{n\,u(n)\} = z^{-1}\cdot\frac{d}{dz^{-1}}\left\{\frac{1}{1-z^{-1}}\right\}$$

$$= \frac{z^{-1}}{(1-z^{-1})^2} \qquad \text{ROC: } |z|>1$$

Applying scaling property

$$z\left\{n\left(-\frac{1}{2}\right)^n u(n)\right\} = \frac{\left\{\left(-\frac{1}{2}\right)^{-1} z\right\}^{-1}}{\left[1-\left\{\left(-\frac{1}{2}\right)^{-1} z\right\}^{-1}\right]^2}$$

$$\therefore X_1(z) = \frac{\left(-\frac{1}{2}\right)z}{\left[1+\frac{1}{2}z^{-1}\right]^2} \qquad \text{ROC: } |z|>\frac{1}{2}$$



For $x_2(n) = \left(\frac{1}{4}\right)^{-n} u(-n) = 4^n u(-n)$

we know,

$$z\{u(n)\} = \frac{1}{1-z^{-1}} \qquad \text{ROC: } |z|>1$$

using Time reversal property

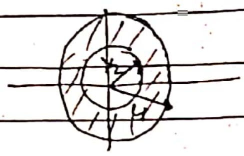
$$z\{u(-n)\} = \frac{1}{1-(z^{-1})^{-1}} = \frac{1}{1-z} \qquad \text{ROC: } |z|<1$$

Applying scaling property.

$\therefore \quad z\{4^n u(-n)\} = \dfrac{1}{1-(4^{-1}z)} = \dfrac{1}{1-\frac{1}{4}z}$; ROC $|z|<4$

Here, i.e. $X_2(z) = \dfrac{4}{4-z}$ ROC : $|z|<4$

$\therefore x(n) = x_1(n) \circledast x_2(n)$



$\therefore X(z) = X_1(z) \cdot X_2(z)$

$\therefore X(z) = \dfrac{(-\frac{1}{2})z^{-1}}{\left[1+\frac{1}{2}z^{-1}\right]^2} \cdot \dfrac{4}{4-z}$ ROC: $\frac{1}{2}<|z|<4$

2014-fall 3. a) Q. Find z-transfrom and its ROC for signal.

$x(n) = \left(\frac{-1}{3}\right)^n u(n) - u(-n-1)$

Soln: Let $x_1(n) = \left(\frac{-1}{3}\right)^n u(n)$

$\therefore$ taking z-transform,

$X_1(z) = \dfrac{1}{1+\frac{1}{3}z^{-1}}$ ROC: $|z|>1/3$ ✓

Let, $x_2(n) = u(-n-1)$

$\therefore X_2(z) = ?$; we know, $z\{u(n)\} = \dfrac{1}{1-z^{-1}}$; $|z|>1$

$u\{(n+1)\}$

Applying time reversal property

$$z\{u(-n)\} = \frac{1}{1-(z^{-1})^{-1}} = \frac{1}{1-z} \quad ROC;\ |z|<1$$

Applying time shifting property.

$$\therefore X_2(z) = Z\{u(-n-1)\} = z^{-1}\left(\frac{1}{1-z}\right) = \quad |z|<1$$

now,

$$x(n) = x_1(n) - x_1(n)$$

$$\therefore X(z) = X_1(z) - X_2(z)$$

$$\therefore X(z) = \frac{1}{1+\frac{1}{3}z^{-1}} - \frac{z^{-1}}{1-z} \quad ROC: \tfrac{1}{3}<|z|<1.$$

OR $u(-n-1)$

$$Z\{u(n)\} = \frac{1}{1-z^{-1}} \quad |z|>1.$$

$$= Z\{u(n+1)\} = \frac{z}{1-z^{-1}} \quad |z|>1$$

$$X_2(z) = Z\{u(-n-1)\} = \frac{z^{-1}}{1-z} \quad |z|<1$$

$$\therefore X(z) = X_1(z) - X_2(z)$$

$$= \frac{1}{1+\frac{1}{3}z^{-1}} - \frac{z^{-1}}{1-z.} \quad ROC: \tfrac{1}{3}<z<1$$

2017-Spring Q. 2.b)

Q: If the causal and stable LTI System is described by following LCCD eq^n

$$3y(n) = y(n-1) + x(n-1)$$

i) Find z-transform transfer function of the system

ii) Find impulse response h(n) of the system

iii) Plot magnitude of this system

Sol^n: Here, $3y(n) = y(n-1) + x(n-1)$

i)⇒ taking z-transform,

$$3Y(z) = z^{-1}Y(z) + z^{-1}X(z)$$

or $Y(z)\{3 - z^{-1}\} = X(z)\{z^{-1}\}$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1}}{3-z^{-1}}$$

∴ z-transform transfer function of the system.

$$H(z) = \frac{z^{-1}}{3-z^{-1}} = \frac{z^{-1}}{3-z^{-1}} \quad ROC: |z^{-1}| < 3;\ |z| > 3$$

i.e $H(z) = \frac{1}{3} \cdot \frac{z^{-1}}{1-\frac{1}{3}z^{-1}}$ ∴ A

ii) taking inverse z-transform of H(z) we get

where, $H(z) = z^{-1} \cdot \frac{1}{3} \cdot \frac{1}{1-\frac{1}{3}z^{-1}}$

$$\therefore h(n) = \frac{1}{3}\left(\frac{1}{3}\right)^{n-1} u(n-1) = \left(\frac{1}{3}\right)^n u(n-1)$$

2016 fall 3(a)

Q. Find the inverse z-transform of the system $X(z) = \dfrac{5-2z^{-1}}{2-z^{-1}-1.5z^{-2}}$

for the ROC: (i) $|z|<1$ (ii) $|z|>3$ (iii) $1<|z|<3$.

Soln:- Here, $X(z) = \dfrac{5-2z^{-1}}{2-z^{-1}-1.5z^{-2}}$

$= \cancel{\dfrac{5-\frac{2}{z}}{2-\frac{1}{z}-1.5\frac{1}{z^2}}}$ $\quad$ or, $X(z) = \dfrac{z^{-1}(5z-2)}{z^{-2}(2z^2-z-1.5)}$

$z = \frac{1}{4} \pm \frac{\sqrt{11}}{4}i$

$X(z) = \dfrac{z(5z-2)}{2z^2-z-1.5}$

$= \cancel{\dfrac{\frac{5z-2}{z}}{\frac{2z^2-z-1.5}{z^2}}}$

Now, dividing both sides by z,

$\dfrac{X(z)}{z} = \dfrac{z(5z-2)}{z(2z^2-z-1.5)}$

$= \cancel{5z-2}$

$\therefore \dfrac{X(z)}{z} = \dfrac{5z-2}{2z^2-z-1.5}$

Let, $\dfrac{X(z)}{z} = 5z-2$

2014 fall Q. 4.b)

An LTI System is characterized by the system function

$$H(z) = \frac{3-4z^{-1}}{1-\frac{7}{2}z^{-1}+\frac{3}{2}z^{-2}}$$

specify ROC of $H(z)$ & find $h(n)$. When the System is stable?

Soln:-

Here, given,

$$H(z) = \frac{3-4z^{-1}}{1-\frac{7}{2}z^{-1}+\frac{3}{2}z^{-2}} = \frac{z^{-1}(3z-4)}{z^{-2}(1\cdot z^{2}-\frac{7}{2}z+\frac{3}{2})}$$

$$\therefore H(z) = \frac{3z-4}{z^{2}-\frac{7}{2}z+\frac{3}{2}}$$

2013 spring

2.b) Q. The z-transform of a particular discrete-time signal $x(n)$ is expressed as $X(z) = \frac{1+\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}$. Determine $x(n)$ using time shifting property.

Soln: Here, given, $X(z) = \frac{1+\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}} = \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}$

$\therefore$ taking inverse z-transform

$x(n) = z^{-1}X(z) = z^{-1}\left[\frac{1}{1-\frac{1}{2}z^{-1}} + \frac{\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}\right]$

or, $x(n) = z^{-1}\left[\frac{1}{1-\frac{1}{2}z^{-1}}\right] + z^{-1}\left[\frac{\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}\right]$

$x(n) = z^{-1}[X_1(z)] + z^{-1}[X_2(z)]$ ——— (1)

where, $X_1(z) = \frac{1}{1-\frac{1}{2}z^{-1}}$ & $X_2(z) = \frac{\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}$

$z^{-1}(X_1(z)) = z\left\{\left(\frac{1}{2}\right)^n . u(n)\right\}$

$X_2(z) = \frac{1}{2} . z^{-1} X_1(z)$

We know

Applying time Shifting Property in $X_2(z)$ i.e. $x(n) \leftrightarrow X(z)$

$x(n-k) \leftrightarrow z^{-k} X(z)$

$z^{-1}.(X_2(z)) = \frac{1}{2}z^{-1}X_1(z) =$

$= \frac{1}{2} z\left\{\left(\frac{1}{2}\right)^{n-1} u(n-1)\right\}$

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Substituting $X_1(z)$ & $X_2(z)$ in eqⁿ (1)

$$x(n) = z^{-1}\left\{z\left(\tfrac{1}{2}\right)^n u(n)\right\} + z^{-1}\left\{\tfrac{1}{2} z\left[u(n-1)\left(\tfrac{1}{2}\right)^{n-1}\right]\right\}$$

$$= \left(\tfrac{1}{2}\right)^n u(n) + \tfrac{1}{2}\left\{u(n-1)\left(\tfrac{1}{2}\right)^{n-1}\right\}$$

$$= \left(\tfrac{1}{2}\right)^n \left\{u(n) + u(n-1)\right\}$$

$$= \left(\tfrac{1}{2}\right)^n \left\{u(n) - u(n-1) + 2u(n-1)\right\}$$

$$\therefore x(n) = \left(\tfrac{1}{2}\right)^n \left\{\delta(n) + 2u(n-1)\right\} \quad \text{Ans}$$

2017 fall Q. 3.a) Given, that $x(n) = a^n u(n)$. determine one-sided z-transform of a signal $x_1(n) = x(n+2)$

Sol: Here given, $x(n) = a^n u(n)$

we know,

$$X(z) = Z\{a^n u(n)\} = \frac{1}{1-az^{-1}} \qquad \text{ROC}: |z| > |a|$$

~~but we have to calculate one-sided z-transform of $x_1(n) = x(n+2)$.~~

since ~~u(n)~~ is one-sided ~~transform~~

$\therefore$ ~~$X_1(z) = Z\{x_1(n)\} = z^2 X(z) = z^2\left(\frac{1}{1-az^{-1}}\right)$ ROC: $|z| > |a|$~~

we know, from time advance property $x(n+k) \xrightarrow{z^+} z^k\left[X^+(z) - \sum_{n=0}^{(k-1)} x(n) z^{-n}\right]$

~~Here, k = 2 so.~~

$$\therefore Z^+[x(n+2)] = z^2\left[X^+(z) - \sum_{n=0}^{2-1} x(n) z^{-n}\right] = z^2 X^+(z) - x(0)z^2 - x(1)z$$

but $x(0) = 1$, $x(1) = a$ and $X^+(z) = \frac{1}{1-az^{-1}}$.

$$\therefore Z^+\{x(n+2)\} = \frac{z^2}{1-az^{-1}} - z^2 - az \quad \text{Ans}$$

2012-fall Q. If the z-transform of any signal is $X(z) = 2+3z^{-1}+4z^{-2}$

3.2) Determine the initial & final values of the Corresponding discrete-time Signal $x(n)$.

2013 spring 3.b)

Soln: Here,

$$X(z) = 2+3z^{-1}+4z^{-2}$$

From initial value theorem

$$x(0) = \lim_{n\to 0} x(n) = \lim_{|z|\to\infty} X(z)$$

$$\therefore x(0) = \lim_{|z|\to\infty} (2+3z^{-1}+4z^{-2}) = \lim_{z\to\infty}\left(2+\frac{3}{z}+\frac{4}{z^2}\right) = 2+\frac{3}{\infty}+\frac{4}{\infty} = 2$$

$$\therefore x(0) = 2+0+0 = 2$$

From final value theorem,

$$x(\infty) = \lim_{n\to\infty} x(n) = \lim_{|z|\to 1}\left[(1-z^{-1})\, X(z)\right]$$

$$\therefore x(\infty) = \lim_{|z|\to 1}\left[(1-z^{-1})(2+3z^{-1}+4z^{-2})\right]$$

$$= \lim_{|z|\to 1}\left[2+2z^{-1}+z^{-2}-4z^{-3}\right]$$

$$= \lim_{|z|\to 1}\left[2+\frac{2}{z}+\frac{1}{z^2}-\frac{4}{z^3}\right]$$

$$= 2+\frac{2}{1}+\frac{1}{1}-\frac{4}{1}$$

$$= 2+2-4$$

$$\therefore x(\infty) = 0$$

$\therefore x(0) = 2$

& $x(\infty) = 0$

2013 Spring 2.b) Q. The z-transform of a particular discrete-time signal $x(n)$ is expressed as $X(z) = \dfrac{1+\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}$

Determine $x(n)$ ~~using time shifting property~~

Soln: Here, $X(z) = \dfrac{1+\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}$

Using differentiation property

$$\frac{dX(z)}{dz^{-1}} = z^{-1}\frac{d}{dz^{-1}}\left\{\frac{1+\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}\right\}$$

$$\frac{dX(z)}{dz^{-1}} = z^{-1}\left\{\frac{\frac{1}{2}\left(1-\frac{1}{2}z^{-1}\right) - \left(-\frac{1}{2}\right)\left(1+\frac{1}{2}z^{-1}\right)}{\left(1-\frac{1}{2}z^{-1}\right)^2}\right\}$$

$$= z^{-1}\frac{\left\{\frac{1}{2} - \frac{1}{4}z^{-1} + \frac{1}{2} + \frac{1}{4}z^{-1}\right\}}{\left(1-\frac{1}{2}z^{-1}\right)^2}$$

$$= z^{-1}\left\{\frac{1+0}{1-\frac{1}{2}z^{-1}}\right\}$$

$$\therefore \frac{dX(z)}{dz^{-1}} = z^{-1}\cdot\frac{1}{1-\frac{1}{2}z^{-1}}$$

multiplying both sides by z^{-1}

$$z^{-1}\frac{dX(z)}{dz^{-1}} = z^{-1}\cdot\left\{\frac{z^{-1}}{1-\frac{1}{2}z^{-1}}\right\}$$

$n\,x(n) = z^{-2}\cdot\left(\frac{1}{2}\right)^n u(n)$

$\therefore n\,x(n) = \left(\frac{1}{2}\right)^n u(n-2)$

$\therefore x(n) = \dfrac{\left(\frac{1}{2}\right)^n u(n-2)}{n}$

2012 spring

Q. Given, $y(n) = 3y(n-1) - 0.1\,y(n-2) + x(n)$

5.2) i) Find the System function, $H(z)$ iv). plot magnitude response

2012-fall ii) is the System is IIR or FIR.

iii) Plote pole-zero diagram.

Soln Here, $y(n) = 3y(n-1) - 0.1\,y(n-2) + x(n)$

taking z-tramform.

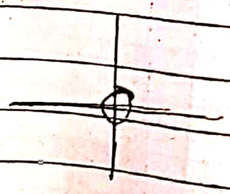
i) ⇒ $Y(z) = 3z^{-1}Y(z) - 0.1\,z^{-2}Y(z) + X(z)$

$\therefore Y(z)\{1 - 3z^{-1} + 0.1\,z^{-2}\} = X(z)$

$\therefore H(z) = \dfrac{Y(z)}{X(z)} = \dfrac{1}{1 - 3z^{-1} + 0.1z^{-2}}$

ii) ⇒ Since, $H(z)$ has. ~~only~~ poles & zero. So, it is IIR System

$H(z) = \dfrac{1}{z^{-2}(z^2 - 3z + 0.1)} = \dfrac{z^2}{z^2 - 3z + 0.1}$



zeros lies at $z = 0$ & ∞.

Poles lies at $z = 2.96$ & 0.0337

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Q. If $X(z) = 2 + 3z^{-1} + 4z^{-2}$ then $x(n) = ?$

Solⁿ: $x(n) = \{2, 3, 4\}$

2013-fall Q. 2.b) By differentiation $X(z)$ and using appropriate properties of z-transform, determine $x(n)$ for the transform given as

$$X(z) = \log(1 - z^{-1}), \quad ROC: |z| < \frac{1}{2} \quad \text{---(1)}$$

Solⁿ:- According to question

differentiate both sides in eqⁿ(1), w.r.t. z^{-1} we get

$$\frac{dX(z)}{dz^{-1}} = \frac{d}{dz^{-1}} \log(1 - z^{-1})$$

$$= \frac{1}{(1 - z^{-1})} \cdot (-1)$$

$$= \frac{-1}{1 - z^{-1}}$$

multiplying both sides by z^{-1} we get

$$z^{-1} \frac{dX(z)}{dz^{-1}} = \cancel{x}\, z^{-1} \left(\frac{-1}{1 - z^{-1}}\right)$$

$$n x(n) = -1 \cdot z^{-1} \cdot \frac{1}{1 - z^{-1}}$$

we know, that

$$z\{u(n)\} = \frac{1}{1-z^{-1}}, \quad ROC: |z^{-1}| < 1 \text{ i.e } |z| > 1$$

so, $n\,x(n) = -z^{-1}u(n) = -u(n-1)$

$\therefore x(n) = \frac{-u(n-1)}{n}$ Ans

2013-Spring 3.a) Q. The input response for a discrete-time ~~signal~~ system is given as $h(n) = \{1, 2, 3\}$ and output response is given as $y(n) = \{1, 1, 2, -1, 3\}$. Determine the $x(n)$ using inverse z-transform.

Soln: Here, given, $h(n) = \{1, 2, 3\}$ then, $H(z) = 1 + 2z^{-1} + 3z^{-2}$

& $y(n) = \{1, 1, 2, -1, 3\} = 1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4}$

then,

$X(z) = ?$ & $x(n) = ?$

• Using long division Method for inverse z-transform

we know, $y(n) = x(n) \otimes h(n)$

$\therefore Y(z) = X(z) \cdot H(z)$

$\therefore X(z) = \dfrac{Y(z)}{H(z)} = \dfrac{1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4}}{1 + 2z^{-1} + 3z^{-2}}$

ie.

$$
\begin{array}{r|l}
 & 1 - z^{-1} + z^{-2} \\
\hline
1 + 2z^{-1} + 3z^{-2} & 1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4} \\
 & -\ 1 + 2z^{-1} + 3z^{-2} \\
 & \quad - z^{-1} - z^{-2} - z^{-3} \\
 & \quad - z^{-1} - 2z^{-2} - 3z^{-3} \\
 & \quad + \quad + \quad + \\
\hline
 & \quad\quad z^{-2} + 2z^{-3} + 3z^{-4} \\
 & \quad\quad z^{-2} + 2z^{-3} + 3z^{-4} \\
 & \quad\quad - \quad - \quad - \\
\hline
 & \quad\quad \times \quad \times \quad \times
\end{array}
$$

$\therefore X(z) = 1 - z^{-1} + z^{-2}$ $\therefore x(n) = \{1, -1, 1\}$

2014 fall Q. 3.b) Find $x(n)$ using inverse z-transform of $X(z) = \dfrac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$ when ROC: $|z| > 1$ and ROC: $|z| < \frac{1}{2}$.

Soln:- i) For ROC: $|z| > 1$

Here, $X(z) = \dfrac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$

$$1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \frac{15}{8}z^{-3} + \frac{3}{c}z^{-4} + \dots$$

$$1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2} \;\Big)\; 1$$

must be in descending order of power for ROC: $|z| > 1$

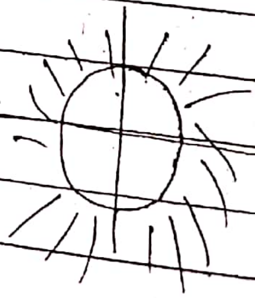
$$1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}$$
$$- \quad + \quad -$$
$$\frac{3}{2}z^{-1} - \frac{1}{2}z^{-2}$$
$$\frac{3}{2}z^{-1} - \frac{9}{4}z^{-2} + \frac{3}{4}z^{-3}$$
$$- \quad + \quad -$$
$$\frac{7}{4}z^{-2} - \frac{3}{4}z^{-3}$$
$$\frac{7}{4}z^{-2} - \frac{21}{8}z^{-3} + \frac{7}{8}z^{-4}$$
$$- \quad + \quad -$$
$$\frac{15}{8}z^{-3} - \frac{7}{8}z^{-4}$$
$$\frac{15}{8}z^{-3} - \frac{45}{16}z^{-4} + \frac{15}{16}z^{-5}$$
$$- \quad + \quad -$$
$$\frac{3}{c}z^{-4} - \frac{15}{16}z^{-5}$$

$\therefore X(z) = 1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \frac{15}{8}z^{-3} + \dots$

taking inverse z-transform. we get

$\therefore$ $x(n) = \{1, \frac{3}{2}, \frac{7}{4}, \frac{15}{8}, \dots\dots\}$ B.

causal(+ve) infinite



ROC: $|z| > 1$

ii) for ROC: $|z| < \frac{1}{2}$.

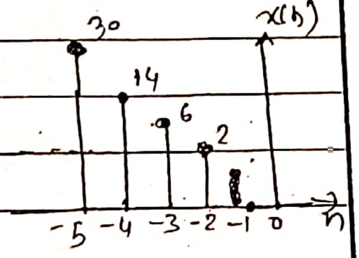
$$
\begin{array}{r|l}
 & 2z^2 + 6z^3 + 14z^4 + 30z^5 + \cdots \\
\frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1 & 1 \\
 & \cancel{2\left(\frac{1}{2}z^{-2} - \frac{3}{2}z^{-1} + 1\right)} \\
 & 1 - 3z + 2z^2 \\
 & - \quad + \quad - \\
 & 3z - 2z^2 \\
 & 3z - 9z^2 + 6z^3 \\
 & - \quad + \quad - \\
 & 7z^2 - 6z^3 \\
 & 7z^2 - 21z^3 + 14z^4 \\
 & - \quad + \quad - \\
 & 15z^3 - 14z^4 \\
 & 15z^3 - 45z^4 + 30z^5 \\
 & - \quad + \quad - \\
 & 36z^4 - 30z^5
\end{array}
$$

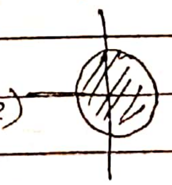
must be in ascending order for ROC: $|z| < \frac{1}{2}$

$\therefore X(z) = 2z^2 + 6z^3 + 14z^4 + 30z^5 + \cdots$

taking inverse of z-transform,

$x(n) = \{ \cdots\cdots 30, 14, 6, 2, 0, \underset{\uparrow}{0} \}$





Anti-causal (-ve) & infinite.

Q: Find the inverse - z-transform of $H(z) = \frac{-4+8z^{-1}}{1+6z^{-1}+8z^{-2}}$

Soln:- By partial fraction method.

Here, $H(z) = \frac{-4+8z^{-1}}{1+6z^{-1}+8z^{-2}} = \frac{-4+8z^{-1}}{(1+2z^{-1})(1+4z^{-1})}$

Let, $\frac{-4+8z^{-1}}{1+6z^{-1}+8z^{-2}} = \frac{-4+8z^{-1}}{(1+2z^{-1})(1+4z^{-1})} = \frac{A}{1+2z^{-1}} + \frac{B}{1+4z^{-1}}$

or, $-4+8z^{-1} = A(1+4z^{-1}) + B(1+2z^{-1})$

or $-4+8z^{-1} = A + 4Az^{-1} + B + 2Bz^{-1}$

or $-4+8z^{-1} = (4A+2B)z^{-1} + (A+B)$

Comparing ~~too~~ equating the coeff. of z^{-1} we get

$4A + 2B = 8$ — (i)

$A + B = -4$ — (ii)

on solving eqn (i) & (ii), we get

$A = 8$ & $B = -12$

$\therefore H(z) = \frac{8}{1+2z^{-1}} - \frac{12}{1+4z^{-1}}$

$\therefore H(z) = 8\left[\frac{1}{1-(-2z^{-1})}\right] - 12\left[\frac{18}{1-(-4z^{-1})}\right]$

taking inverse - z - transform.

$\therefore h(n) = 8(2)^n u(n) - 12(-4)^n u(n)$

Q. Find inverse z-transform of $X(z) = \dfrac{z^3}{(z+1)(z-1)^2}$ i.e $x(n) = ?$

Soln: By partial fraction method:

Here, $X(z) = \dfrac{z^3}{(z+1)(z-1)^2}$

Here, the power of num. & den. are same so divide both sides by z. we get

$$\frac{X(z)}{z} = \frac{1}{z} \cdot \frac{z^3}{(z+1)(z-1)^2} = \frac{z^2}{(z+1)(z-1)^2}$$

Let, $$\frac{z^2}{(z+1)(z-1)^2} = \frac{A}{z+1} + \frac{B}{z-1} + \frac{C}{(z-1)^2}$$

or, $z^2 = A(z-1)^2 + B(z+1)(z-1) + C(z+1)$

or $z^2 = Az^2 - 2Az + A + Bz^2 - B + Cz + C$

or $z^2 = (A+B)z^2 + (-2A+C)z + (A+C-B)$

Equating the coeff. of z we get

$A + B = 1$ ——— (I)

$-2A + C = 0$ ——— (II)

$A - B + C = 0$ ——— (III)

on solving we get

$A = \frac{1}{4}$ $B = \frac{3}{4}$ $C = \frac{1}{2}$

$$\therefore \frac{X(z)}{z} = \frac{1/4}{z+1} + \frac{3/4}{z-1} + \frac{1/2}{(z-1)^2}$$

$$\therefore X(z) = \frac{(1/4)z}{z+1} + \frac{(3/4)z}{z-1} + \frac{(1/2)z}{(z-1)^2}$$

taking inverse z-tansform,

$$x(n) = \frac{1}{4}(-1)^n u(n) + \frac{3}{4}(1)^n u(n) + \frac{1}{4}(1)^n n.u(n)$$

2015-Spring 3.a) Q. Determine the inverse z-transform of

$$X(z) = \frac{z}{3z^2-4z+1} \quad \text{if ROC are}$$

i) $|z| > 1$ (ii) $|z| < \frac{1}{3}$ (iii) $\frac{1}{3} < |z| < 1$

Soln: use partial fraction method not long division method

Here, $X(z) = \frac{z}{3z^2-4z+1}$

(always) Dividing both sides by 'z' we get

$$\frac{X(z)}{z} = \frac{z}{z(3z^2-4z+1)}$$

$$= \frac{1}{3z^2-4z+1}$$

$$\frac{X(z)}{z} = \frac{1}{3z^2-3z-z+1} = \frac{1}{(z-1)(3z-1)}$$

Let, $$\frac{1}{(z-1)(3z-1)} = \frac{A}{z-1} + \frac{1}{3z-1}$$

$$A = \left.\frac{1}{3z-1}\right|_{z=1} = \frac{1}{2}$$

$$B = \left.\frac{1}{z-1}\right|_{z=\frac{1}{3}} = \frac{-3}{2}$$

$$\therefore \frac{X(z)}{z} = \frac{1/2}{z-1} - \frac{3/2}{3z-1}$$

$$\therefore X(z) = \frac{1}{2}\left(\frac{z}{z-1}\right) - \frac{1}{2}\frac{z}{\left(z-\frac{1}{3}\right)}$$

i) For $|z| > 1$.

$$x(n) = \frac{1}{2}(1)^n u(n) - \frac{1}{2}\left(\frac{1}{3}\right)^n u(n)$$

causal   causal

ii) For $|z| < \frac{1}{3}$

$$x(n) = \frac{1}{2}(1)^n . u(-n-1) - \frac{1}{2}\left(\frac{1}{3}\right)^n u(-n-1)$$

or $$x(n) = \underbrace{\frac{1}{2}(1)^{-n-1} u(-n-1) - \frac{1}{2}\left(\frac{1}{3}\right)^{n-1}}_{\text{both non-causal}} u(-n-1)$$

iii) For $\frac{1}{3} < |z| < 1$

$$x(n) = \frac{1}{2}(1)^n u(-n-1) \quad - \frac{1}{2}\left(\frac{1}{3}\right)^n u(n)$$

$\uparrow$ non-causal $\qquad$ $\uparrow$ causal.

2016-Spring

Q. The z-transform of particular discrete signal $x(n)$ is expressed as

$X(z) = \frac{1+\frac{1}{2}z^{-1}}{1-\frac{1}{2}z^{-1}}$. Determine $x(n)$

Solⁿ:- By using long division method.

$$1-\frac{1}{2}z^{-1} \overline{)\; 1+\frac{1}{2}z^{-1} \;}\; 1 + z^{-1} + \frac{1}{2}z^{-2} + \dots$$

$$\frac{-\left(1 - \frac{1}{2}z^{-1}\right)}{z^{-1}}$$

$$\frac{-\left(z^{-1} - \frac{1}{2}z^{-2}\right)}{\frac{1}{2}z^{-2}}$$

$$\frac{-\left(\frac{1}{2}z^{-2} - \frac{1}{4}z^{-3}\right)}{\frac{1}{4}z^{-3}}$$

∴ $X(z) =$